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Roll No

EC-704 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Microwave Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the principle of operation of an isolator.
2. Describe the operation of a 2-hole directional coupler and obtain scattering matrix for it and discuss some applications of directional couplers.
3. What is a rectangular waveguide? Derive the wave equation for TE wave and obtain all the field components in a rectangular waveguide.
4. An air filled, rectangular waveguide of $b = 1\text{cm}$ is supposed to operate at 12GHz with propagation constant 150rad/m . What should 'a' in order to support the TE_{10} mode?
5. What is standing wave? Derive the equation of Voltage Standing Wave Ratio. (VSWR) in terms of reflection coefficient.
6. With the help of two valley model along with the emphasis of drift velocity, explain the negative resistance property of a Gunn diode.
7. Write short notes on the following:
 - a) IMPATT
 - b) TRAPATT
8. Write short notes (any four)
 - a) Network analyzer
 - b) TEM mode of operation
 - c) LASER
 - d) Slow wave structure
 - e) Working of reflex klystron
